

Software Testing Assessment Report

Project: Agro India — Digital Agricultural Platform & E-Commerce

Target Platform:	Agro India Web & Mobile App	Evaluation Focus:	Test Strategy, CRUD & Risk Analysis
Domain:	Agri-Tech & Input Supply	Release Status:	Pre-Production Release Candidate

Part 1: Test Case Design

Structured test scenarios and cases addressing regional onboarding, product catalogs, pricing limits, and boundary compliance for the Agro India platform.

1. User Registration & Farmer Profiles

TC_REG_01	Farmer Mobile / Email Onboarding (Positive)	Register a new farmer account using a valid 10-digit mobile number or email, verifying OTP/password authentication and successful profile record creation.
TC_REG_02	Malformed Contact Information (Negative)	Attempt registration with a malformed phone number (fewer than 10 digits) or invalid email structure and verify inline validation blocks submission.
TC_REG_03	Regional Multilingual Language Preference (Edge)	Register while selecting a regional Indian language preference (e.g., Hindi, Telugu, Punjabi) and verify that the UI updates instantly upon activation.

2. User Login & Role-Based Access

TC_LOG_01	Role-Based Dashboard Redirection (Positive)	Log in with verified credentials and ensure redirection to the appropriate dashboard based on assigned user role (Farmer, Distributor, or Admin).
TC_LOG_02	Unverified Credentials Handling (Negative)	Attempt login with incorrect passwords or unverified phone numbers and verify that authentication fails safely with generic alerts.

TC_LOG_03 Inactive Session Timeout Protection (Edge)

Verify that inactive user sessions expire securely after a set threshold to protect sensitive account and transaction profiles.

3. Agro-Input / Product CRUD Operations

TC_CRUD_01 Agro-Input Catalog Creation (Positive)

Add a new agricultural product (Bio-pesticides, Fertilizers, Seeds) with valid name, category, pricing, stock quantity, and imagery; verify catalog indexing.

TC_CRUD_02 Negative Pricing and Missing SKU Validation (Negative)

Attempt to create a product listing with a negative price or missing mandatory SKU fields and verify the database rejects the entry.

TC_CRUD_03 Catalog Search & Category Filtration (Positive)

Browse and filter product catalogs by categories (Plant Growth Regulators, Organic Carbon) or search keywords to ensure accurate query retrieval.

TC_CRUD_04 Inventory Stock & Price Adjustment (Update)

Modify stock levels or adjust pricing for an agro-product and verify updates reflect instantly for front-end buyers.

TC_CRUD_05 Discontinued Product Removal (Delete)

Remove a discontinued product or crop advisory record and ensure it is safely soft-deleted or removed from active public listings.

4. Input Validation & Agricultural Specifics

TC_VAL_01 Inventory Stock Bounds Verification (Negative)

Order an amount of fertilizer exceeding available inventory stock limits and verify that checkout flags a stock-shortage error.

TC_VAL_02 Regional Character Search Injection (Edge)

Input regional characters or special symbols into the product or crop disease search bar to ensure robust text sanitization and no server crashes.

Part 2: Bug Identification & Risk Areas

Potential high-risk failure modes identified for pre-production evaluation without executing the application:

Bug / Risk Description	Severity	Reason & Operational Impact
Inaccurate Floating-Point Calculations	Major	Agricultural items are sold in bulk quantities and fractional weights. Rounding errors can corrupt invoice totals for large farmer orders.
Broken Object Level Authorization (BOLA)	Critical	Weak authorization checks could allow registered farmers to view or alter another farmer's order history or invoices by modifying URL IDs.
Multilingual Text Overflows	Minor	Rendering Indian regional text can trigger layout wrap issues or overlapping buttons on mobile browsers if Unicode fonts aren't bounded.
Inventory Race Conditions	Critical	Concurrent purchase attempts for limited seasonal stock can permit duplicate checkouts, causing negative inventory states and fulfillment failure.
Unthrottled OTP Gateway Requests	Major	Lack of rate limiting on SMS requests leaves endpoints vulnerable to spam scripts, depleting API credits and locking out genuine users.
Unsanitized Crop Disease Uploads	Critical	Permitting file uploads without rigorous extension checks enables attackers to execute malicious scripts disguised as image formats.
Timezone Desynchronization	Minor	Server time mismatches can cause flash sales, weather alerts, or time-sensitive harvest advisory schedules to display erroneous local hours.
SQL Injection in Search Filters	Critical	Inadequately parameterized query strings used for searching fertilizers or inputs can expose the underlying database to total compromise.